

Name: _____

Date: _____

Class: _____

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

Earlier on (in K3, if you completed it) you provided general lessons learned related to engineering design, and now (in L2) you are asked to provide recommendations for continuing work on this project in particular.

Answer as many of these questions as you can with specific recommendations based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design specific decisions you made). Provide recommendations to someone doing a similar project that might build or continue upon your work. For example: "Taking into account <something we did>, we believe doing _____ would improve _____ because _____."

- What could be done next to go further?

The new microphone holder design could be improved through being printed from different materials: a plastic injection mold printer would be able to mass produce the product, and make them of better and stronger quality. A more advanced securing method could be adapted to improve on velcro, as velcro is more subjected to human error in the initial application process, as well as in everyday use. Also, different types of holders could be designed to hold different types of microphone transmitters, such as a Bose XLR transmitter or a Godox MoveLink II transmitter.

1. A production manufacturing company would have advanced factory systems to mass produce the product in plastic injection printers
2. With a more advanced printer, a more securing method such as a slide and lock system could be designed and implemented.
3. The design could be modified to change the slot for the transmitter to fit whichever transmitter size and shape is used.

- What would you keep if you did the project again?

If we did the project again, we would definitely keep the prototype design we had of a removable head for the holder assembly. It protects the microphone transmitter well, and looks authentic. We would also keep the relative design of the holder.

1. Keep the detachable head design
2. Alter it slightly to look more authentic or be secured a different way

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3. Add a foam grip to the holder shaft for comfort to hold

- What would you change?

We considered replacing velcro with a slide and locking mechanism for the securing method. We were unable to make this with the resources available to us, so if we were equipped with better technology such as a plastic injection printer we would be able to make a more advanced securing method. The slot for the transmitter could also be altered to fit different types of transmitters.

1. Add a slide and lock design to secure the head on the holder
2. Mass print on a plastic injection printer
3. Change slot for transmitter dependent on type.
 - What would have made this project easier for you to complete?

This project would be easier to complete if we had a different printer, such as a plastic injection printer. We were able to come up with a good design, so there was not much we needed to make the project easier. The biggest flaw in the process was the number of failed prints we had, as the screw on threads design completely failed, and even our latest design failed to print twice.

1. Acquire a more advanced printer (could be through a stakeholder company)
2. Increasingly durable materials until a design is essentially immune to breaking.

In order to earn 5's on Element L, you will need specific and detailed ways the project can be improved and a rationale for why they are needed.

Detailed Plans for Implementation of Improvements (L2)

Now that you have generated a list of specific recommendations with their rationale, go back under each and generate a consistently detailed list about HOW these details will be implemented. Refer back to Element I where you concluded with some of these ideas.

1. Use plastic injection printer to make better 3D prints
2. Make multiple slot designs to fit different mic transmitters
3. Make a more advanced securing method using more advanced technology
4. Use more durable materials
5. Make head textured to look like a real microphone head